

HIGH LEVEL LANGUAGE EXTENSION WITH CURSORS

Problem- 1

Write a Cursor program using MySQL to retrieve the email-ids(build an email list) of employees from employees table.

```
mysql> INSERT INTO employees VALUES
-> (1, 'Harry Potter', 'pharry@warnerbros.com'),
-> (2, 'Clark Kent', 'kclark@dccomics.com'),
-> (3, 'Tony Stark', 'stony@marvel.com');
```

Query OK, 3 rows affected (0.03 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM employees;
```

id	Name	email
1	Harry Potter	pharry@warnerbros.com
2	Clark Kent	kclark@dccomics.com
3	Tony Stark	stony@marvel.com

3 rows in set (0.00 sec)

```
mysql> DELIMITER $$
```

```
mysql> CREATE PROCEDURE build_email_list(INOUT email_list VARCHAR(4000))
```

```
-> BEGIN
```

```
->     DECLARE v_finished INTEGER DEFAULT 0;
```

```
->     DECLARE v_email VARCHAR(100) DEFAULT '';
```

```
->
```

```
->     DECLARE email_cursor CURSOR FOR
```

```
->         SELECT email FROM employees;
```

```
->
```

```
->     DECLARE CONTINUE HANDLER FOR NOT FOUND
```

```
->         SET v_finished = 1;
```

```
->
```

```
->     OPEN email_cursor;
```

```
->
```

```
->     get_email: LOOP
```

```
->         FETCH email_cursor INTO v_email;
```

```
->
```

```
->         IF v_finished = 1 THEN
```

```
->             LEAVE get_email;
```

```
->         END IF;
```

```
->
```

```
->         SET email_list = CONCAT(v_email, ';', email_list);
```

```
->     END LOOP get_email;
```

```
->
```

```
->     CLOSE email_cursor;
```

```
-> END$$
```

Query OK, 0 rows affected (0.37 sec)

```
mysql> DELIMITER ;
mysql> SET @email_list = '';
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> CALL build_email_list(@email_list);
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT @email_list;
+-----+
| @email_list |
+-----+
| stony@marvel.com;kclark@dccomics.com;pharry@warnerbros.com; |
+-----+
1 row in set (0.00 sec)
```

```
mysql> CALL build_email_list(@email_list);
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT @email_list;
+-----+
| @email_list |
+-----+
| stony@marvel.com;kclark@dccomics.com;pharry@warnerbros.com;stony@marvel.com;kclark@dccomics.com;pharry@warnerbros.com; |
+-----+
1 row in set (0.00 sec)
```